



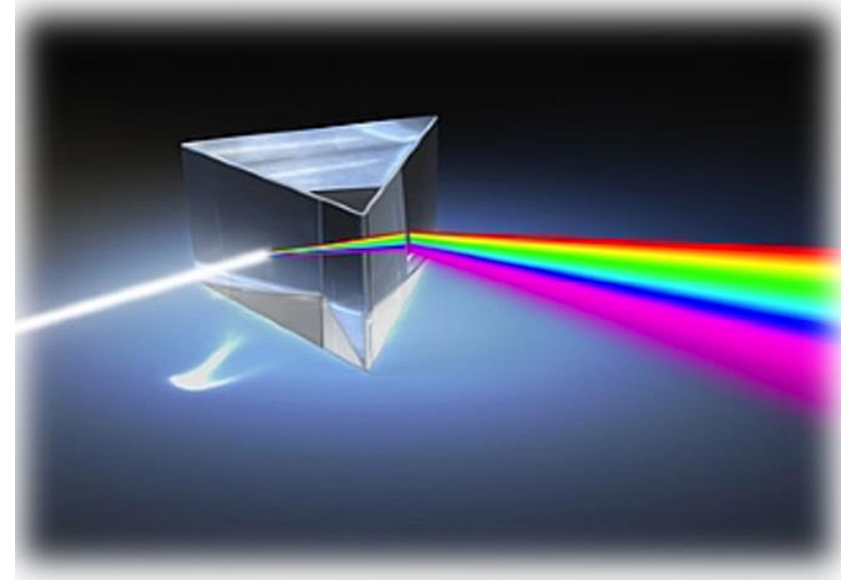
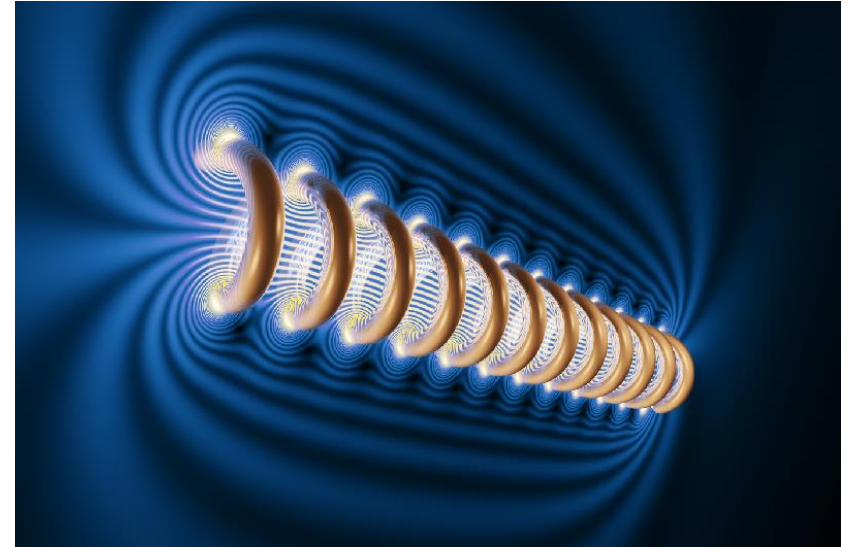
Density Functional Perturbation Theory and its Time-Dependent Extension

Jacques K. Desmarais



Outline

- Inclusion of Static Electric Fields for Molecules
- The Coupled-Perturbed Kohn-Sham Procedure
- Infinite Periodic Systems: the Modern Theories
- Dynamic Fields: TD-KS
- Linear Response in the Frequency Domain



Inclusion of Static Electric Fields

- Because the HK theorems, as well as the KS procedure work for any scalar-multiplicative v_{ext} , we can also take: $\hat{v}_{ext} = \hat{v}_{n-e} + \hat{v}_{field}$
- with v_{field} being the potential that describes a static external electric field
- In general, the static electric field is a wave oscillating with momentum $\mathbf{q}$, obtained as a solution of Maxwell's equations (*for molecules*):

$$v_{field} = i\mathcal{E} \cdot \frac{\mathbf{q}}{q^2} e^{i\mathbf{q} \cdot \mathbf{r}} = i\mathcal{E} \cdot \frac{\mathbf{q}}{q^2} (1 + i\mathbf{q} \cdot \mathbf{r} + \dots)$$

- In many experiments, the wavelength of the field is much larger than the dimensions of the molecule so we take $\mathbf{q} \rightarrow \mathbf{0}$ (spatially homogeneous field):

$$v_{field} = -\mathcal{E} \cdot \mathbf{r}$$

- This is the *dipole approximation*. Higher-order terms (quadrupole, octupole, ...) can be used in short wavelength experiments (e.g. hard X-rays)



Properties as Energy Derivatives

$$\frac{\partial^{n_R+n_\epsilon+n_B+n_{I_i}} E(\mathbf{R}, \epsilon, \mathbf{B}, \mathbf{I}_i)}{\partial \mathbf{R}^{n_R} \partial \epsilon^{n_\epsilon} \partial \mathbf{B}^{n_B} \partial \mathbf{I}_i^{n_{I_i}}}$$

Table 3. Response Properties of the Potential Energy

n_R	n_ϵ	n_B	n_{I_i}	property
0	0	0	0	energy
1	0	0	0	forces
2	0	0	0	Hessian (harmonic frequencies)
0	1	0	0	dipole moment (IR)
1	1	0	0	infrared absorption intensities (IR)
0	2	0	0	polarizability (Raman)
0	0	1	1	nuclear magnetic shielding (NMR)
0	0	0	2	nuclear spin–spin coupling (NMR)
0	1	1	0	optical rotation (circular dichroism)

Keith et al., 2021



Properties as Energy Derivatives

Now let's calculate $\partial E / \partial \mathcal{E} |_{\mathcal{E}=0}$ and $\partial^2 E / \partial \mathcal{E}^2 |_{\mathcal{E}=0}$!



Calculating Energy Derivatives: CPKS/DFPT

- We want to solve the problem: $\hat{H}|i\rangle = \epsilon_i|i\rangle$
- with the Hamiltonian: $\hat{H} = -\frac{1}{2}\nabla^2 + \hat{v}_{ext} + \hat{v}_{field} + \hat{v}_{Hxc}$
- We suppose that we have already solved without $v_{field} = -\mathcal{E} \cdot \mathbf{r}$:

$$\hat{H}^{(0)}|i^{(0)}\rangle = \left(-\frac{1}{2}\nabla^2 + \hat{v}_{ext} + \hat{v}_{Hxc}^{(0)}[n^{(0)}]\right)|i^{(0)}\rangle = \epsilon_i^{(0)}|i^{(0)}\rangle$$
- We can use this zeroth-order solution by expanding everything in powers of $\mathcal{E}$:

$$n = n^{(0)} + \mathcal{E}n^{(1)} + \mathcal{E}^2n^{(2)} + \dots$$

$$\hat{v}_{Hxc} = \hat{v}_{Hxc}^{(0)}[n^{(0)}] + \mathcal{E}\hat{v}_{Hxc}^{(1)}[n^{(0)}, n^{(1)}] + \mathcal{E}^2\hat{v}_{Hxc}^{(2)}[n^{(0)}, n^{(1)}, n^{(2)}] + \dots$$

$$\hat{H} = \hat{H}^{(0)} + \mathcal{E}\hat{H}^{(1)} + \mathcal{E}^2\hat{H}^{(2)} + \dots$$

$$|i\rangle = |i^{(0)}\rangle + \mathcal{E}|i^{(1)}\rangle + \mathcal{E}^2|i^{(2)}\rangle + \dots$$

$$\epsilon_i = \epsilon_i^{(0)} + \mathcal{E}\epsilon_i^{(1)} + \mathcal{E}^2\epsilon_i^{(2)} + \dots$$

Calculating Energy Derivatives: CPKS/DFPT

- Which leads to the equations:

$$\hat{H}^{(0)}|i^{(0)}\rangle = \epsilon_i^{(0)}|i^{(0)}\rangle$$

$$\hat{H}^{(1)}|i^{(0)}\rangle + \hat{H}^{(0)}|i^{(1)}\rangle = \epsilon_i^{(1)}|i^{(0)}\rangle + \epsilon_i^{(0)}|i^{(1)}\rangle$$

$$\hat{H}^{(2)}|i^{(0)}\rangle + \hat{H}^{(1)}|i^{(1)}\rangle + \hat{H}^{(0)}|i^{(2)}\rangle = \epsilon_i^{(2)}|i^{(0)}\rangle + \epsilon_i^{(1)}|i^{(1)}\rangle + \epsilon_i^{(0)}|i^{(2)}\rangle$$

- In CRYSTAL we expand the orbitals in Nth order in an atomic orbital basis: $\dots$

$$|i^{(N)}\rangle = \sum_{\mu} c_{\mu,i}^{(N)} |\mu\rangle$$

- Then, defining the matrices $H_{\mu\nu}^{(N)} = \langle\mu|\hat{H}^{(N)}|\nu\rangle$ and $S_{\mu\nu} = \langle\mu|\nu\rangle$ and left-multiplying the above equations by $\langle\mu|$ gives the first-order matrix equation:

$$\mathbf{H}^{(1)}\mathbf{c}^{(0)} + \mathbf{H}^{(0)}\mathbf{c}^{(1)} = \mathbf{S}\mathbf{c}^{(0)}\epsilon^{(1)} + \mathbf{S}\mathbf{c}^{(1)}\epsilon^{(0)}$$



Calculating Energy Derivatives: CPKS/DFPT

- We can analytically solve the matrix equation:

$$\mathbf{H}^{(1)}\mathbf{c}^{(0)} + \mathbf{H}^{(0)}\mathbf{c}^{(1)} = \mathbf{S}\mathbf{c}^{(0)}\epsilon^{(1)} + \mathbf{S}\mathbf{c}^{(1)}\epsilon^{(0)}$$

- By introducing the auxiliary $\mathbf{G}^{(1)}$ and $\mathbf{U}^{(1)}$ matrices:

$$\mathbf{G}^{(1)} = [\mathbf{c}^{(0)}]^\dagger \mathbf{H}^{(1)} \mathbf{c}^{(0)} \quad \text{and} \quad \mathbf{c}^{(1)} = \mathbf{c}^{(0)} \mathbf{U}^{(1)}$$

- What does $\mathbf{U}^{(1)}$ like to eat? Does it prefer wine or beer? To find out, we can also use the orthonormality condition: $\langle j|i \rangle = \delta_{ij} \rightarrow \mathbf{c}^\dagger \mathbf{S} \mathbf{c} = \mathbf{1}$ which, expanded in orders of $\mathcal{E}$ gives:

$$[\mathbf{c}^{(0)}]^\dagger \mathbf{S} \mathbf{c}^{(0)} = \mathbf{1}$$

$$[\mathbf{c}^{(1)}]^\dagger \mathbf{S} \mathbf{c}^{(0)} + [\mathbf{c}^{(0)}]^\dagger \mathbf{S} \mathbf{c}^{(1)} = \mathbf{0}$$

...



Calculating Energy Derivatives: CPKS/DFPT

- We can solve:

$$\left[\mathbf{c}^{(1)}\right]^{\dagger} \mathbf{S} \mathbf{c}^{(0)} + \left[\mathbf{c}^{(0)}\right]^{\dagger} \mathbf{S} \mathbf{c}^{(1)} = \mathbf{0}$$

- By inserting $\mathbf{c}^{(1)} = \mathbf{c}^{(0)} \mathbf{U}^{(1)}$ and using also zeroth-order orthonormality:

$$\left[\mathbf{U}^{(1)}\right]^{\dagger} + \mathbf{U}^{(1)} = \mathbf{0}$$

- Using this result and left-multiplying the first-order KS equation by $\left[\mathbf{c}^{(0)}\right]^{\dagger}$ gives:

$$\mathbf{G}^{(1)} + \epsilon^{(0)} \mathbf{U}^{(1)} = \epsilon^{(1)} + \epsilon^{(0)} \mathbf{U}^{(1)}$$

- which is the first-order CPKS/DFPT equation!



Calculating Energy Derivatives: CPKS/DFPT

- A certain level of arbitrariness exists in the solution of the 1st order CPKS/DFPT equation, because $\mathbf{U}^{(1)}$ can be anything we want, so long as it is a solution of:

$$\left[\mathbf{U}^{(1)}\right]^\dagger + \mathbf{U}^{(1)} = \mathbf{0} \quad \text{and} \quad \mathbf{G}^{(1)} + \epsilon^{(0)}\mathbf{U}^{(1)} = \epsilon^{(1)} + \epsilon^{(0)}\mathbf{U}^{(1)}$$

- The typical procedure is to let the occ-occ and virt-virt blocks of $\mathbf{U}^{(1)}$ be Hermitian, which leads to:

$$\mathbf{U}_{OO}^{(1)} = \mathbf{U}_{VV}^{(1)} = \mathbf{0} \quad \text{and} \quad \begin{aligned} \epsilon_{OO}^{(1)} &= \mathbf{G}_{OO}^{(1)} \\ \epsilon_{VV}^{(1)} &= \mathbf{G}_{VV}^{(1)} \end{aligned}$$

- Finally, using $\epsilon_{OV}^{(1)} = \mathbf{0}$ gives:

$$\mathbf{U}_{OV}^{(1)} = - \left[\mathbf{U}_{VO}^{(1)}\right]^\dagger = \frac{\mathbf{G}_{OV}^{(1)}}{\epsilon_V^{(0)} - \epsilon_O^{(0)}}$$



Calculating Energy Derivatives: CPKS/DFPT

- With $\mathbf{U}^{(1)}$ having been determined, we can immediately calculate the first order energy (electric polarization $\mathbf{P}$):

$$E^{(1)} = \left. \frac{\partial E}{\partial \mathcal{E}} \right|_{\mathcal{E}=0} = - \sum_i \langle \varphi_i | \mathbf{r} | \varphi_i \rangle$$

- The procedure can also be extended to 2nd order (electric polarizability α and dielectric constant ϵ^∞):

$$E^{(2)} = \left. \frac{\partial^2 E}{\partial \mathcal{E}^2} \right|_{\mathcal{E}=0} = -\text{Tr} \left\{ [\mathbf{r}]_{OV} \mathbf{U}_{VO}^{(1)} \right\}$$

- ...and so on. In general, it is possible to calculate up to $E^{(3)}$ using only $\mathbf{U}^{(1)}$. This is a particular result of a theorem in perturbation theory called the 2n+1 rule



Calculating Energy Derivatives: CPKS/DFPT

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* *      REFERENCES TO BE QUOTED WHEN USING THIS MODULE:      * *
* *      (1) M. Ferrero, M. Rerat, R. Orlando, R.Dovesi,      * *
* *      J. Comput. Chem. 29 (2008), 1450;      * *
* *      J. Chem. Phys. 128 (2008), 014110.      * *
* *      (2) M. Ferrero, M. Rerat, B. Kirtman, R. Dovesi,      * *
* *      J. Chem. Phys. 129 (2008), 244110.      * *
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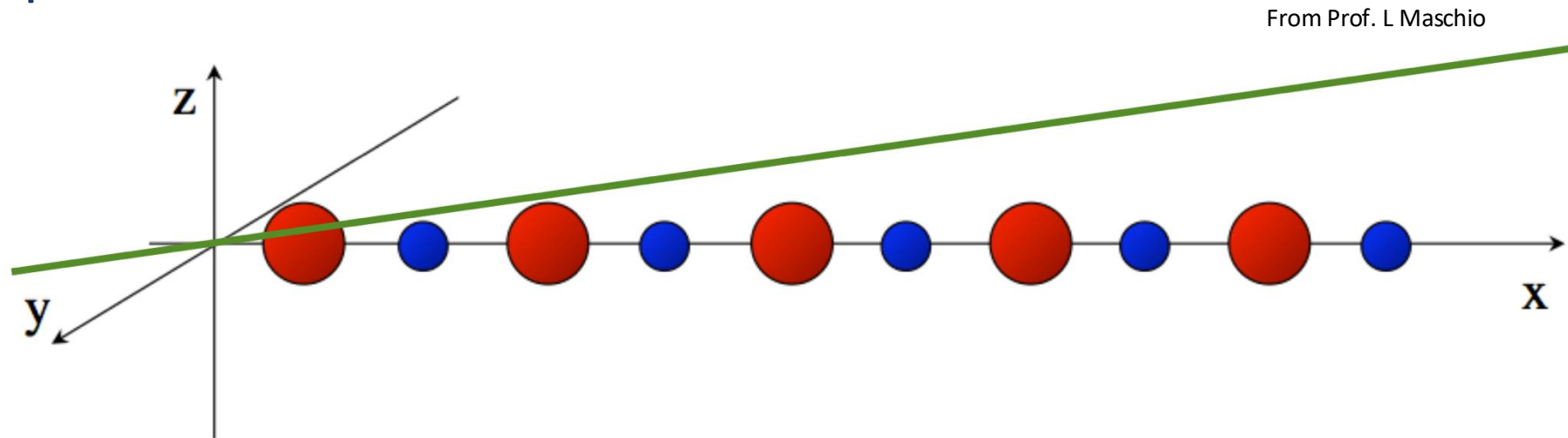
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TT				CP MONMON	TELAPSE	6727.87	TCPU	6660.26	

Periodic Systems: The Modern Theories

- Coming back to static electric fields, the operator:

$$v_{\text{field}} = -\mathcal{E} \cdot \mathbf{r}$$

- Is problematic for calculations with conventional Bloch functions, because it is not lattice periodic!



Periodic Systems: The Modern Theories

- Then, $\mathbf{P}$ is determined by charges at the edges of the sample: $\mathbf{P} = Q_{end}$

$$Q_{end} = \frac{\gamma_C}{2\pi} = \frac{i}{2\pi} \oint \langle u_{\mathbf{k}} | \partial_{\mathbf{k}} u_{\mathbf{k}} \rangle \cdot d\mathbf{k} \in \mathbb{Z}$$

PHYSICAL REVIEW B

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Theory of polarization of crystalline solids

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(Received 10 June 1992)

We consider the change in polarization $\Delta\mathbf{P}$ which occurs upon making an adiabatic change in the Kohn-Sham Hamiltonian of the solid. A simple expression for $\Delta\mathbf{P}$ is derived in terms of the valence-band wave functions of the initial and final Hamiltonians. We show that physically $\Delta\mathbf{P}$ can be interpreted as a displacement of the center of charge of the Wannier functions. The formulation is successfully applied to compute the piezoelectric tensor of GaAs in a first-principles pseudopotential calculation.

Macroscopic polarization in crystalline dielectrics: the geometric phase approach

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The macroscopic electric polarization of a crystal is often defined as the dipole of a unit cell. In fact, such a dipole moment is ill defined, and the above definition is incorrect. Looking more closely, the quantity generally measured is *differential* polarization, defined with respect to a "reference state" of the same material. Such differential polarizations include either derivatives of the polarization (dielectric permittivity, Born effective charges, piezoelectricity, pyroelectricity) or finite differences (ferroelectricity). On the theoretical side, the differential concept is basic as

Quantal phase factors accompanying adiabatic changes

BY M. V. BERRY, F.R.S.

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Tyndall Avenue, Bristol BS8 1TL, U.K.*

(Received 13 June 1983)

A quantal system in an eigenstate, slowly transported round a circuit C by varying parameters $\mathbf{R}$ in its Hamiltonian $\hat{H}(\mathbf{R})$, will acquire a geometrical phase factor $\exp\{i\gamma(C)\}$ in addition to the familiar dynamical phase factor. An explicit general formula for $\gamma(C)$ is derived in terms of the



Dynamic Fields: TD-KS scheme

- Much as in the time-independent case, it is possible to map the interacting problem to a non-interacting system that reproduces the exact time-dependent density
- This allows to write a time-dependent KS equation:

$$\left(-\frac{1}{2}\nabla^2 + v_{ext}(\mathbf{r}, t) + \int d^3r' \frac{n(\mathbf{r}', t)}{|\mathbf{r}-\mathbf{r}'|} + v_{xc}(\mathbf{r}, t) \right) \varphi_i(\mathbf{r}, t) = i \frac{\partial}{\partial t} \varphi_i(\mathbf{r}, t)$$

- Most practical applications use the potential of static DFT and plug in the time-dependent density, which is inserted in the TDKS equation. This is the so-called *adiabatic approximation*:

$$v_{xc}^A(\mathbf{r}, t) = v_{xc}^0[n_0](\mathbf{r}) \Big|_{n_0(\mathbf{r}) \rightarrow n(\mathbf{r}, t)}$$



Linear Response in the Frequency Domain

- Involves solving a non-Hermitian matrix equation. Which, in the adiabatic approx, can be solved by a Sternheimer equation approach (i.e. dynamic CPKS)

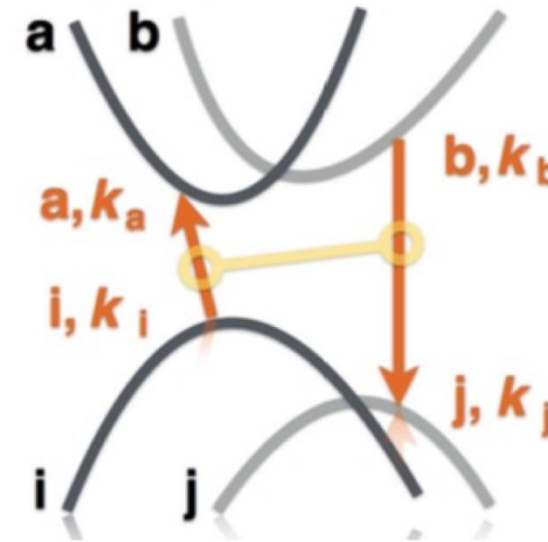
$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^* & \mathbf{A}^* \end{pmatrix} \begin{pmatrix} \mathbf{X} \\ \mathbf{Y} \end{pmatrix} = \omega \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \mathbf{X} \\ \mathbf{Y} \end{pmatrix}$$

$\omega \rightarrow$ transition energies

$|\mathbf{X}, \mathbf{Y}\rangle \rightarrow$ intensities

$$\begin{aligned} A_{ai,bj}^{k_a k_i, k_b k_j} = & \delta_{ij} \delta_{ab} \delta_{k_i k_j} \delta_{k_a k_b} (\varepsilon_a^{k_a} - \varepsilon_i^{k_i}) \\ & + (a k_a i k_i | j k_j b k_b) \\ & - (a k_a b k_b | j k_j i k_i)_{CHF} \\ & + (a k_a i k_i | f_{xc}^{AL} | j k_j b k_b) \end{aligned}$$

$$\begin{aligned} B_{ai,bj}^{k_a k_i, k_b k_j} = & (a k_a i k_i | b k_b j k_j) \\ & - (a k_a j k_j | b k_b i k_i)_{CHF} \\ & + (a k_a i k_i | f_{xc}^{AL} | b k_b j k_j) \end{aligned}$$



From Bernasconi,
MSSC2022

CRYSTAL References

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Periodic Systems: The Modern Theories

- For infinite, periodic systems, the wave function has translational symmetry:

$$|\psi(\mathbf{r} + N\mathbf{g})|^2 = |\psi(\mathbf{r})|^2$$

- This leads to:

$$\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r} + i\phi(\mathbf{k})} u_{\mathbf{k}}(\mathbf{r})$$

- With cell periodic functions:

$$u_{\mathbf{k}}(\mathbf{r} + \mathbf{g}) = u_{\mathbf{k}}(\mathbf{r})$$

$$H\psi_{\mathbf{k}} = E_{\mathbf{k}}\psi_{\mathbf{k}}$$



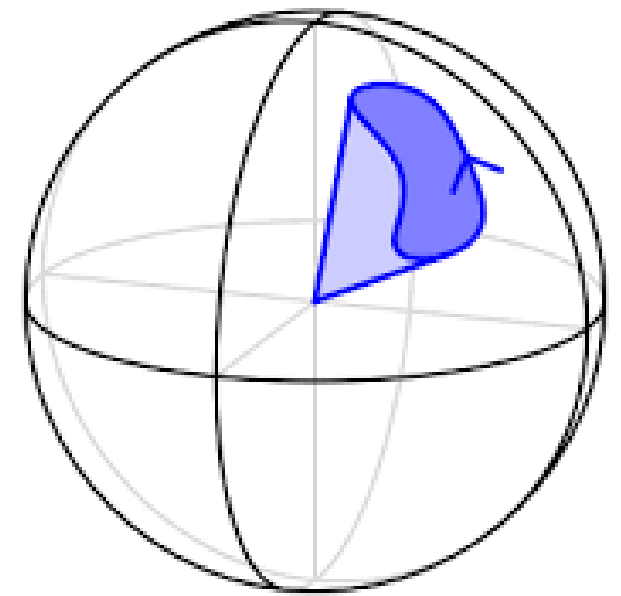
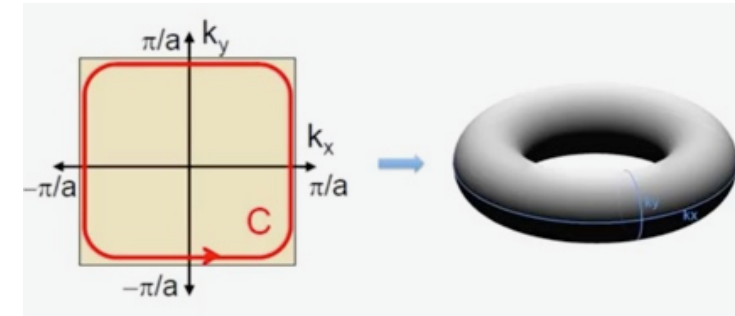
Periodic Systems: The Modern Theories

- Then, $\mathbf{P}$ is determined by charges at the edges of the sample:
- Recall that the wave function includes an arbitrary phase ϕ :

$$\psi'_k(\mathbf{r}) = e^{i\phi(\mathbf{k})} \psi_k(\mathbf{r}) \rightarrow |\psi'_k|^2 = |\psi_k|^2$$

- The system undergoes a cycle $\mathbf{k} \rightarrow \mathbf{k} + \mathbf{K}$, which is adiabatic (or sufficiently slow, in the sense that the system remains in an eigenstate of the Hamiltonian)
- Then the state remains unchanged, but the wave function can, in general, differ by a phase
- For a contour C , we can write $e^{i\phi} = e^{i\theta + i\gamma_C}$, where γ_C is the part of the phase associated with C

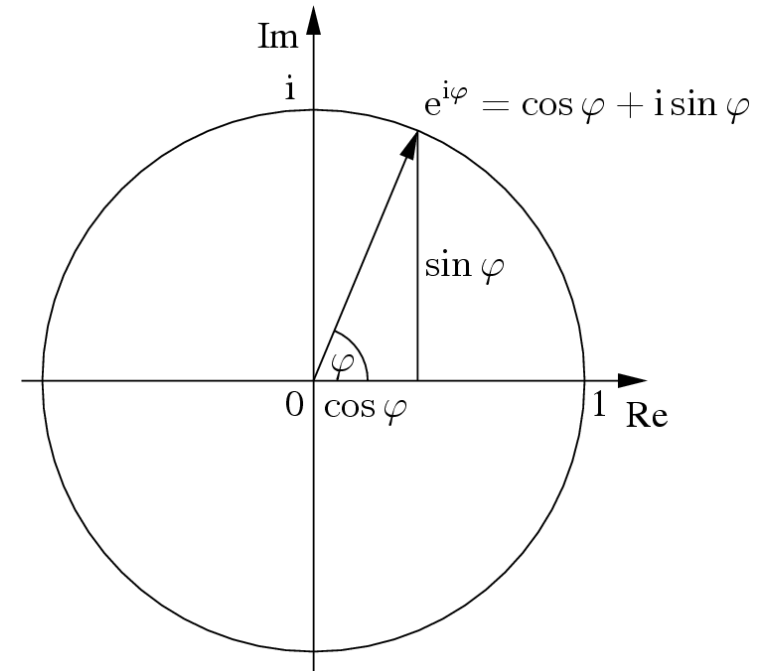
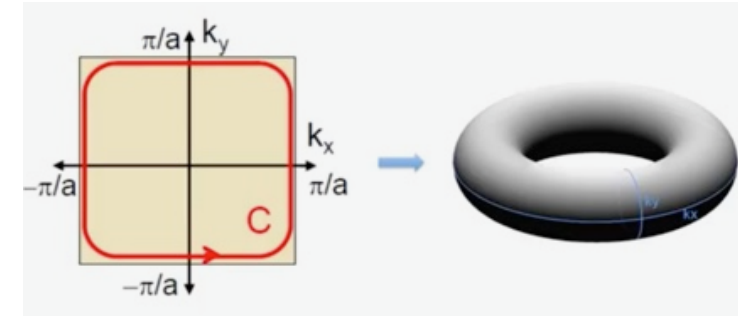
$$\mathbf{P} = Q_{end}$$



Periodic Systems: The Modern Theories

- Then, $\mathbf{P}$ is determined by charges at the edges of the sample:
- For a contour C , we can write $e^{i\phi} = e^{i\theta + i\gamma_C}$, where γ_C is the part of the phase associated with C
- $\gamma_C = \oint_C \mathbf{A} \cdot d\mathbf{k}$ and then we can determine $\mathbf{A}$ by the requirement that $|\Psi(t)\rangle$ remains a solution of the Schrodinger equation: $\mathbf{A} = i\langle u_{\mathbf{k}} | \partial_{\mathbf{k}} u_{\mathbf{k}} \rangle$
- Notice that $e^{i\gamma_C} = 1$, for $\gamma_C = 2\pi n$, where $n \in \mathbb{Z}$
- So for a closed path: $\gamma_C = 2\pi n, n \in \mathbb{Z}$

$$\mathbf{P} = Q_{end}$$



Runge-Gross Theorems

- The HK theorems provide a reformulation of the *time-independent* Schrodinger equation $H \Psi = E \Psi$
- The RG theorems instead provide a reformulation of the *time-dependent* Schrodinger equation $H(t) \Psi(t) = i \frac{\partial}{\partial t} \Psi(t)$, which is necessary for studying the effects of dynamic (time-oscillating) fields, with $H(t) = T + V_{ee} + V_{ne} + V_{\text{field}}(t)$
- The theorem (not shown here) proceeds as:

$$v(\mathbf{r}, t) = v[n, \Psi_0](\mathbf{r}, t) \implies \hat{H}(t) = \hat{H}[n, \Psi_0](t) \implies \Psi(t) = \Psi[n, \Psi_0](t) .$$

- Which means that we can calculate all observables as functionals of the time-dependent density:

$$O(t) = \langle \Psi[n, \Psi_0] | \hat{O}(t) | \Psi[n, \Psi_0] \rangle = O[n, \Psi_0](t)$$



Runge



Gross



Inclusion of Static Magnetic Fields: CSDFT

- A magnetic field is not scalar-multiplicative, so the regular HK theorems do not work:

$$\mathbf{B} = \nabla \times \mathbf{A}$$

- The magnetic field enters the Hamiltonian through the Zeeman interaction and through the vector-potential as $\mathbf{p} \rightarrow \mathbf{p} + \frac{1}{c} \mathbf{A}$:

$$E = \langle \Psi | \frac{1}{2} \left[i \nabla + \frac{1}{c} \mathbf{A} \right]^2 + \hat{V}_{ee} + \hat{V}_{ne} | \Psi \rangle + \beta_e \int \mathbf{B}(\mathbf{r}) \cdot \mathbf{m}(\mathbf{r}) d\mathbf{r}$$

- An integration by parts gives:

$$E = \langle \Psi | \hat{T} + \hat{V}_{ee} | \Psi \rangle + \int \left[\hat{v}_{n-e} + \frac{1}{2c^2} \mathbf{A} \cdot \mathbf{A} \right] n(\mathbf{r}) d\mathbf{r} + \frac{1}{c} \int \mathbf{j}(\mathbf{r}) \cdot \mathbf{A}(\mathbf{r}) d\mathbf{r} + \beta_e \int \mathbf{B}(\mathbf{r}) \cdot \mathbf{m}(\mathbf{r}) d\mathbf{r}$$

- Then, Vignale and Rasolt (1987) showed that E is a unique functional of $n, \mathbf{m}$ and $\mathbf{j}$: $E = E[n, \mathbf{m}, \mathbf{j}]$ this is called current-spin DFT
- $\mathbf{j}$ is the velocity field of n (the particle current)



M. Rasolt



G. Vignale



Inclusion of Static Magnetic Fields: CSDFT

- The same coupled-perturbed procedure (with small modifications) can be used to calculate the orbital response to magnetic fields, rather than electric fields
- For magnetic fields, assuming, as before, that the wavelength of the field is much larger than the dimension of the molecule (spatially homogeneous field) gives:

$$v_{\text{field}} = -\mathbf{B}_0 \cdot (\mathbf{r} \times \mathbf{p})$$

- And, then, for example, the first-order energy (orbital magnetization):

$$E^{(1)} = \sum_i \langle \varphi_i | v_{\text{field}} | \varphi_i \rangle$$

- For magnetic fields, the formulation is more involved beyond first order



Periodic Systems: The Modern Theories

- In the mid 2000's a similar development has provided a formula for the orbital magnetization:

$$\mathbf{M} = \frac{1}{2c} \Im \int d^3k \sum_i f_i \langle \partial_{\mathbf{k}} u_{i,\mathbf{k}} | \times (H_{\mathbf{k}} - \epsilon_{i,\mathbf{k}} - 2\epsilon_F) | \partial_{\mathbf{k}} u_{i,\mathbf{k}} \rangle$$

- where $H_{\mathbf{k}} = e^{-i\mathbf{k}\cdot\mathbf{r}} H e^{i\mathbf{k}\cdot\mathbf{r}}$, f_i are fractional occupations and ϵ_F is the Fermi energy

PRL **99**, 197202 (2007)

PHYSICAL REVIEW LETTERS

week ending
9 NOVEMBER 2007

Quantum Theory of Orbital Magnetization and Its Generalization to Interacting Systems

Junren Shi,¹ G. Vignale,² Di Xiao,³ and Qian Niu³

Periodic Systems: The Modern Theories

- Then, $\mathbf{P}$ is determined by charges at the edges of the sample:
- Closed path around the BZ: $\gamma_C = 2\pi Q_{end}$

$$\mathbf{P} = Q_{end}$$

